Riphah International University

Gulberg Campus

OOP



Submitted by

ZEENAT FAYYAZ

Sap id

78482

Submitted to

Miss Nayab Khalid

Lab 6

Task 1

A Team has many Players, but each Player can exist without the Team (he can join another team).

- Create a Player class with jerseyNo and name.
- Create a Team class with a list of Players.
- Deleting the Team must NOT delete the Players.
- In main, create some Players, add them to the Team, and display them.

Code

```
import java.util.ArrayList; import
java.util.List;

class Player {
private int jerseyNo;
private String name;

   public Player(int jerseyNo, String name) {
this.jerseyNo = jerseyNo;         this.name =
name;
   }

   public String getName() {
return name;
   }

   public int getJerseyNo() {
return jerseyNo;
   }

   @Override     public String toString() {         return "Player [Jersey
No: " + jerseyNo + ", Name: " + name + "]";
   }
}

class Team {     private String
teamName;     private
List<Player> players;
```

```
    public Team(String teamName) {
this.teamName = teamName;         this.players
= new ArrayList<>();
    }

    public void addPlayer(Player player) {
players.add(player);
    }

    public void displayTeam() {
        System.out.println("\n--- " + teamName + " Roster ---");
if (players.isEmpty()) {
            System.out.println("No players in this team.");
        } else {
            for (Player p : players) {
                System.out.println(p);
            }
        }
    }
}

public class Main {     public static void
main(String[] args) {

        Player p1 = new Player(10, "Lionel Messi");
        Player p2 = new Player(7, "Cristiano Ronaldo");
        Player p3 = new Player(11, "Neymar Jr");

        Team myTeam = new Team("All-Stars FC");
myTeam.addPlayer(p1);         myTeam.addPlayer(p2);
myTeam.addPlayer(p3);

        myTeam.displayTeam();

        System.out.println("\n[System] Disbanding the team...");
myTeam = null; // The team object is now destroyed

        System.out.println("\n--- Checking Independent Players ---");
System.out.println(p1);
        System.out.println(p2);
        System.out.println(p3);
    }
}
```

Output

```
Output


--- All-Stars FC Roster ---
Player [Jersey No: 10, Name: Lionel Messi]
Player [Jersey No: 7, Name: Cristiano Ronaldo]
Player [Jersey No: 11, Name: Neymar Jr]

[System] Disbanding the team...

--- Checking Independent Players ---
Player [Jersey No: 10, Name: Lionel Messi]
Player [Jersey No: 7, Name: Cristiano Ronaldo]
Player [Jersey No: 11, Name: Neymar Jr]
```

Task 2

A Car has an Engine. The Engine is built inside the Car and cannot exist without it.

- Create an Engine class with a start() method that prints a message.
- Create a Car class that makes its own Engine inside the constructor.
- In main, create a Car and call drive(), which starts the Engine.

Code

```
class Engine {
public void start() {
        System.out.println("Engine started: Vroom Vroom!");
    }
}

class Car {     private String model;     private Engine engine; // The
Engine cannot exist without the Car      public Car(String model) {
this.model = model;

        this.engine = new Engine();
    }

    public void drive() {
```

```
        System.out.println("Preparing to drive the " + model + "...");
engine.start(); // Invoking the engine's start method
        System.out.println("The car is now moving.");
    }
}

public class Main {
    public static void main(String[] args) {
Car myCar = new Car("Tesla Model S");
myCar.drive();
    }
}
```

Output

